Data Inventory and Data Dictionary

Python Programming II

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Research question

Who rides Boston's fare-free bus routes and how did ridership and service performance on these routes change over the years compared with similar routes that kept fares?

Data Storage

Since the file size is too large, we have kept them in Google Drive in their raw format. After we clean the sheets up and extract useful information, we will add them to GitHub Repository.

Google Drive link -
https://drive.google.com/drive/folders/1UkYo4iAItXL2bdpykE8rIO_X56W3cSUi?usp=sharing

Data Files and Sources

We are using tabular data available in csv format published by the Boston Transit Authority or under the American Community Census.

1. American Community Survey 5-Year Estimates, 2020–2024 | *U.S. Census Bureau, ACS Data API, census.gov/data/developers/data-sets/acs-5year.html*
2. TIGER/Line Census Tract Boundaries, 2024 | *U.S. Census Bureau, TIGER/Line Shapefiles, https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html*
3. MBTA GTFS Static Feed | *MBTA Developers, mbta.com/developers/gtfs*
4. MBTA Bus Arrival/Departure Times | *MBTA Blue Book Open Data Portal, mbta-massdot.opendata.arcgis.com*
5. MBTA Bus Ridership by Trip, Season, Route/Line, and Stop | *MBTA Blue Book Open Data Portal, mbta-massdot.opendata.arcgis.com*
6. MBTA Systemwide Passenger Survey 2015-17 | *Central Transportation Planning Staff (CTPS), Boston Region MPO, for the MBTA, https://www.ctps.org/dv/mbtasurvey2018/index.html*
7. MBTA 2025 System-Wide Passenger Survey | *MBTA Office of Performance Management and Innovation, via Massachusetts GIS Data Hub, https://gis.data.mass.gov/datasets/MassDOT::mbta-2025-system-wide-passenger-survey*
8. Fare-Free Bus Program Mid-Program Report | *City of Boston, Streets Cabinet, https://www.boston.gov/departments/transportation/free-route-23-28-and-29-bus-program*

Data dictionary

American Community Survey 5-Year Estimates, 2020–2024

Variable	Description
NAME	Tract name, county and state
B19013_001E	Median household income, inflation-adjusted dollars
B17001_001E	Population for whom poverty status is determined
B17001_002E	Population with income below the poverty level
B08201_001E	Total households
B08201_002E	Households with no vehicle available
B03002_001E	Total population
B03002_003E	Not Hispanic or Latino, White alone
B03002_004E	Not Hispanic or Latino, Black or African American alone
B03002_006E	Not Hispanic or Latino, Asian alone
B03002_012E	Hispanic or Latino (any race)
B08301_001E	Workers 16 and over (means of transportation to work)
B08301_010E	Workers commuting by public transportation (excluding taxicab)
B08303_001E	Workers 16 and over who did not work from home (travel time universe)
B08303_011E	Travel time to work: 45 to 59 minutes
B08303_012E	Travel time to work: 60 to 89 minutes
B08303_013E	Travel time to work: 90 or more minutes
B25003_001E	Occupied housing units
B25003_003E	Renter-occupied housing units
B25070_001E	Renter-occupied units (gross rent as a share of income universe)
B25070_007E	Gross rent is 30.0 to 34.9% of household income
B25070_008E	Gross rent is 35.0 to 39.9% of household income
B25070_009E	Gross rent is 40.0 to 49.9% of household income
B25070_010E	Gross rent is 50.0% or more of household income
C16002_001E	Total households (household language universe)
C16002_004E	Limited English-speaking households: Spanish
C16002_007E	Limited English-speaking households: other Indo-European languages
C16002_010E	Limited English-speaking households: Asian and Pacific Island languages
C16002_013E	Limited English-speaking households: other languages
B23025_003E	Civilian labor force, population 16 and over
B23025_005E	Civilian labor force: unemployed
state	State FIPS code
county	County FIPS code
tract	Tract code

TIGER/Line Census Tract Boundaries, 2024

Field	Description
STATEFP	State FIPS code
COUNTYFP	County FIPS code
TRACTCE	Tract code
GEOID	State + county + tract
GEOIDFQ	Fully qualified GEOID used by Census web services
NAME	Tract number as a short label
NAMELSAD	Tract name with legal description
MTFCC	Feature class code (G5020 = census tract)
FUNCSTAT	Functional status code
ALAND	Land area (square meters)
AWATER	Water area (square meters)
INTPTLAT, INTPTLON	Latitude and longitude of the tract's internal point
geometry	Tract boundary polygon

MBTA GTFS Static Feed

Table	Key fields
routes.txt	route_id, route_short_name, route_long_name, route_type
stops.txt	stop_id, stop_name, stop_lat, stop_lon, parent_station, wheelchair_boarding
trips.txt	trip_id, route_id, service_id, direction_id, shape_id
stop_times.txt	trip_id, stop_id, stop_sequence, arrival_time, departure_time, timepoint
shapes.txt	shape_id, shape_pt_lat, shape_pt_lon, shape_pt_sequence
calendar.txt	service_id, weekday flags, start_date, end_date

MBTA Bus Arrival/Departure Times

Variable	Description
service_date	Date of service
route_id	Bus route identifier
direction_id	Direction of travel
half_trip_id	Identifier for one one-way trip
stop_id	Stop identifier (matches GTFS)
time_point_id	Timepoint code (a stop where schedule adherence is measured)
time_point_order	Order of the timepoint along the trip
point_type	Position on trip: start point, midpoint or end point
standard_type	Whether the trip is evaluated against a schedule or a headway standard
scheduled	Scheduled arrival or departure time
actual	Actual arrival or departure time recorded by vehicle tracking
scheduled_headway	Scheduled gap from the previous bus
headway	Actual gap from the previous bus

MBTA Bus Ridership by Trip, Season, Route/Line, and Stop

Variable	Description
GTFS route_id	Route identifier
GTFS direction_id	Direction of travel
trip start time	Scheduled start time of the trip
Stop Name	Stop name
GTFS stop_id	Stop identifier
stop sequence	Order of the stop on the trip
Year	Year of the Fall season measured
Day Type	Weekday, Saturday or Sunday
Boardings	Average passengers boarding at the stop on this trip
Alightings	Average passengers getting off at the stop on this trip
Load	Average passengers on board when leaving the stop
Route/Variant	Route variant (branch or pattern) the trip belongs to
# of Trip Samples	Number of observed trips averaged to produce the values

MBTA Systemwide Passenger Survey, 2015–17

Sheet	Contents
Line values	Weighted rider estimates by line
Line percentages	Same rows, as shares of respondents
Station values	Weighted rider estimates by station
Station percentages	Same rows, as shares

MBTA 2025 System-Wide Passenger Survey

Variable	Description
aggregation_level	Level of the row: Reporting Group, Rapid Transit or Bus Rapid Transit Line, Service Mode, or Systemwide
service_mode	Mode, e.g. Bus, Commuter Rail, a named rapid transit line
reporting_group	Route, route group or area
measure_group	Topic (9): Access Mode, Alternative Modes, Fares, Income, Race and Ethnicity, Smartphones, Trip Purpose and Frequency, Vehicles, Other Demographics
measure	Question
category	Answer option
weighted_percent	Weighted share of riders giving this answer
n	Number of survey responses in the reporting group for this measure
margin_of_error	Margin of error for weighted_percent
ObjectId	Row identifier added by the data portal

Fare-Free Bus Program Mid-Program Report

A PDF report from the City of Boston summarizing findings from rider surveys, focus groups and MBTA data.